

ALGORITHM :- SUPPORT VECTOR MACHINE

Support Vector Machine (SVM) is a supervised machine learning algorithm mainly used for classification (and also regression).

It works by finding the best decision boundary (called a hyperplane) that separates classes in the feature space with the maximum possible margin.

Algorithm type & Objective

- **Algorithm Type:**
 - Margin-based classifier – finds a separating hyperplane between classes with maximum margin.
 - Supervised learning – trained on labeled data (reviews with ratings).
 - Can be linear (LinearSVC) or non-linear (kernel SVM).
- **Objective:**
 - For binary classification: Find the hyperplane that maximizes the margin between two classes.
 - For multi-class classification: Use One-vs-Rest (OvR) or One-vs-One (OvO) strategies to extend binary SVM.
 - In review rating: Assign the most probable rating (1–5 stars) based on learned decision boundaries.

How the Algorithm Works (Mathematical Intuition / Workflow)

1. Feature Representation

- Convert text reviews into numerical vectors (e.g., TF-IDF, Bag of Words).

2. Find Optimal Hyperplane

- A hyperplane is defined by:

$$w \cdot x + b = 0$$

- The goal is to maximize the margin (distance between the hyperplane and nearest points of each class).

3. Optimization Problem

- Hard-margin SVM (perfectly separable data):

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

$$y_i(w \cdot x_i + b) \geq 1$$

- Soft-margin SVM (real-world noisy reviews):

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i$$

Where:

- C = penalty for misclassification
- ξ_i = slack variables allowing margin violations

4. Kernel Trick for Non-linearity

- Replace $x \cdot x'$ with a **kernel function** $K(x, x')$ to implicitly map data to a higher-dimensional space.

5. Decision Rule

- Predict:

$$\hat{y} = \text{sign}(w \cdot x + b)$$

- For multi-class: pick the class with the highest decision function score.

Key Hyperparameters

Hyperparameter	What It Controls	Default	Tuning Tips
C	Regularization strength (trade-off between margin size and misclassification)	1.0	Smaller values = wider margin but more misclassifications; larger values = narrower margin, less tolerance for errors.

Hyperparameter	What It Controls	Default	Tuning Tips
kernel	Type of transformation (linear, poly, rbf, sigmoid)	rbf	Use linear for high-dimensional text data; rbf for non-linear cases.
gamma	Influence of a single training example (RBF/poly kernels)	scale	Lower = smoother decision boundary; higher = more complex fit.
degree	Degree of polynomial kernel	3	Only relevant if kernel='poly'.
max_iter	Max number of iterations	-1 (no limit)	Increase if convergence issues occur.

Strengths & Limitations

Strengths

- Works well with high-dimensional sparse text data like TF-IDF.
- Can model non-linear decision boundaries using kernels.
- Robust to overfitting when regularized properly.
- Good for small-to-medium datasets.

Limitations

- Slow to train on very large datasets.
- Kernel SVMs are memory-intensive for large vocabularies.
- Not inherently probabilistic (needs extra computation for probability outputs).
- Less interpretable compared to logistic regression.

When to Use / When Not to Use

Use When:

- You have high-dimensional text data and need strong classification performance.
- You have a small-to-medium dataset and can afford longer training times.
- You need to capture non-linear patterns between words and labels.

Avoid When:

- Dataset is very large (training and prediction will be slow).
- You need a fully interpretable model for feature importance.

- You need real-time predictions on massive streams of reviews.